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Semiconductor arrangement including first and second gate electrodes and method of manufacture

Abstract

A method of forming a semiconductor arrangement includes forming a gate dielectric layer over a semiconductor layer. A gate electrode layer is formed over the gate dielectric layer. A first gate mask is formed over the gate electrode layer. The gate electrode layer is etched using the first gate mask as an etch template to form a first gate electrode. A first dopant is implanted into the semiconductor layer using the first gate mask and the first gate electrode as an implantation template to form a first doped region in the semiconductor layer.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5904528	12/1998	Lin et al.	N/A	N/A
5972783	12/1998	Arai et al.	N/A	N/A
6008077	12/1998	Maeda	N/A	N/A
6063547	12/1999	Ye et al.	N/A	N/A
6225151	12/2000	Gardner et al.	N/A	N/A
6596599	12/2002	Guo	N/A	N/A
6794279	12/2003	Stephen et al.	N/A	N/A
10840333	12/2019	Wu et al.	N/A	N/A
11575008	12/2022	Wu et al.	N/A	N/A
2007/0290192	12/2006	Rotondaro	257/19	H01L 21/823412
2014/0246696	12/2013	Flachowsky	257/190	H01L 29/6659

OTHER PUBLICATIONS

U.S. Appl. No. 62/753,152 Drawings. cited by examiner

U.S. Appl. No. 62/753,152 Specification. cited by examiner

Tian-I Liou, et al., "Hot-Electron-Induced Degradation of Conventional, Minimum Overlap, LDD

and DDD N-Channel MOSFETs”, JIEEE Circuits and Devices Magazine, Mar. 1988, pp. 9-15. cited by applicant

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of and claims priority to U.S. patent application Ser. No. 18/106,001, titled “SEMICONDUCTOR ARRANGEMENT AND METHOD OF MANUFACTURE” and filed on Feb. 6, 2023, which is a continuation of and claims priority to U.S. patent application Ser. No. 17/098,751, titled “SEMICONDUCTOR ARRANGEMENT AND METHOD OF MANUFACTURE” and filed on Nov. 16, 2020, which is a continuation of and claims priority to U.S. patent application Ser. No. 16/371,535, titled “SEMICONDUCTOR ARRANGEMENT AND METHOD OF MANUFACTURE” and filed on Apr. 1, 2019, which claims priority to U.S. Provisional Application 62/753,152, titled “SEMICONDUCTOR ARRANGEMENT AND METHOD OF MANUFACTURE” and filed on Oct. 31, 2018. U.S. patent application Ser. No. 18/106,001, U.S. patent application Ser. No. 17/098,751, U.S. patent application Ser. No. 16/371,535, and U.S. Provisional Application 62/753,152 are incorporated herein by reference.

BACKGROUND

(1) In a semiconductor device, such as a transistor, current flows through a channel region between a source region and a drain region upon application of a sufficient voltage or bias to a gate of the device. The gate is isolated from the channel region by a gate dielectric layer. When current flows through the channel region, the device is generally regarded as being in an ‘on’ state, and when current is not flowing through the channel region, the device is generally regarded as being in an ‘off’ state. As technology nodes shrink, in some integrated circuit designs, the gate structure height and width dimensions also shrink. Low doped drain (LDD) regions defined in the source/drain regions reduce the likelihood of hot carrier injection that may potentially damage the gate dielectric layer as the height and width dimensions shrink.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIGS. 1-16 are illustrations of a semiconductor arrangement at various stages of fabrication, in accordance with some embodiments.

DETAILED DESCRIPTION

(3) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and

second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(4) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(5) One or more techniques for fabricating a semiconductor arrangement are provided herein. In some embodiments, the semiconductor arrangement comprises a transistor. In some embodiments, an etch process for forming a gate structure and an implantation process for forming a doped region adjacent the gate structure are performed using a patterned mask stack. In some embodiments, a sidewall spacer is formed adjacent the gate structure, and a second dopant is implanted using the first sidewall spacer and the first gate structure as an implantation template to form a second doped region in the semiconductor layer. In some embodiments, a depth of the first doped region is lower than a depth of the second doped region relative to a top surface of the semiconductor layer.

(6) Turning to FIG. 1, a plurality of layers used in the formation of a semiconductor arrangement **100** are illustrated, in accordance with some embodiments. FIG. 1 includes a plan view showing where various cross-sectional views are taken. Referring to FIG. 1, the view X1-X1 is a cross-sectional view taken through the semiconductor arrangement **100** in a direction corresponding to a gate length direction through a first active region **102A**, and the view X2-X2 is a cross-sectional view taken through the semiconductor arrangement **100** in a direction corresponding to a gate length direction through a second active region **102B**. Not all aspects of the processing shown in the cross-sectional views will be depicted in the plan view.

(7) In some embodiments, the semiconductor arrangement **100** comprises transistors. The plurality of layers is formed over a semiconductor layer **105**. In some embodiments, the semiconductor layer **105** is part of a substrate comprising at least one of an epitaxial layer, a single crystalline semiconductor material such as, but not limited to Si, Ge, SiGe, InGaAs, GaAs, InSb, GaP, GaSb, InAlAs, GaSbP, GaAsSb, and InP, a silicon-on-insulator (SOI) structure, a wafer, or a die formed from a wafer. In some embodiments, the semiconductor layer **105** comprises crystalline silicon. In some embodiments, well regions are formed in the semiconductor layer **105** to define the first active region **102A** and the second active region **102B**. In some embodiments, the well regions are of different conductivity types. It will be appreciated that in the cross-sectional views, the well regions are not shown to simplify the drawings.

(8) In some embodiments, a gate dielectric layer **110** is formed over the semiconductor layer **105**. In some embodiments, the gate dielectric layer **110** comprises silicon and oxygen. According to some embodiments, the gate dielectric layer **110** is formed by at least one of thermal oxidation, chemical oxidation, chemical vapor deposition (CVD), including low pressure CVD (LPCVD), plasma enhanced CVD (PECVD), ultra-high vacuum CVD (UHVCVD), reduced pressure CVD (RPCVD), atomic layer deposition (ALD), physical vapor deposition, pulsed laser deposition, sputtering, evaporative deposition, vapor phase epitaxy (VPE), molecular beam epitaxy (MBE), liquid phase epitaxy (LPE), or other applicable techniques.

(9) In some embodiments, a gate electrode layer **115** is formed over the gate dielectric layer **110** by at least one of CVD, LPCVD, PECVD, UHVCVD, RPCVD, ALD, physical vapor deposition, pulsed laser deposition, sputtering, evaporative deposition, VPE, MBE, LPE, or other applicable

techniques. In some embodiments, the gate electrode layer **115** comprises polysilicon.

(10) In some embodiments, a first mask layer **120** is formed over the gate electrode layer **115**. According to some embodiments, the first mask layer **120** comprises a plurality of individually formed layers that together define a first mask stack. In some embodiments, the first mask layer **120** comprises a hard mask layer **125** formed over the gate electrode layer **115** by at least one of CVD, LPCVD, PECVD, UHVCVD, RPCVD, ALD, physical vapor deposition, pulsed laser deposition, sputtering, evaporative deposition, VPE, MBE, LPE, or other applicable techniques. In some embodiments, the hard mask layer **125** comprises silicon and nitrogen.

(11) In some embodiments, the first mask layer **120** comprises a bottom antireflective coating (BARC) layer **130** formed over the hard mask layer **125**. In some embodiments, the BARC layer **130** is a polymer layer that is applied using a spin coating process. In some embodiments, an organic planarization layer (OPL) **135** is formed over the BARC layer **130**. In some embodiments, the OPL layer **135** comprises a photo-sensitive organic polymer that is applied using a spin coating process. In some embodiments, the OPL layer **135** comprises a dielectric layer.

(12) In some embodiments, the first mask layer **120** comprises a second hard mask layer **137** formed over a region of the semiconductor arrangement **100** comprising the second active region **102B** while not being formed over the first active region **102A**. In some embodiments, the second hard mask layer **137** is formed over the first active region **102A** and the second active region **102B**, a mask layer is formed over the second active region **102B**, an etch process is performed to remove the second hard mask layer **137** over the first active region **102A**, and then the mask layer formed over the second active region **102B** is removed.

(13) According to some embodiments, the first mask layer **120** comprises a photoresist layer **140** formed over the OPL layer **135** and the second hard mask layer **137**. In some embodiments, the photoresist layer **140** is formed by at least one of spinning, spray coating, or other applicable techniques. The photoresist layer **140** comprises an electromagnetic radiation sensitive material and properties, such as solubility, of the photoresist layer **140** are affected by electromagnetic radiation. The photoresist layer **140** is either a negative photoresist or a positive photoresist. In some embodiments, portions of the OPL layer **135** are irradiated by the electromagnetic radiation that patterns the photoresist layer **140** are affected to change the etch selectivity of the irradiated portions of the OPL layer **135** with respect to non-irradiated portions.

(14) Referring to FIG. 2, the first mask layer **120** is patterned to define a first gate mask **122**, in accordance with some embodiments. In some embodiments, the photoresist layer **140** is exposed using a radiation source and a reticle to define a pattern in the photoresist layer **140**. In some embodiments, portions of the photoresist layer **140** are removed to define a patterned photoresist layer **142**. In some embodiments, the underlying OPL layer **135**, BARC layer **130**, and hard mask layer **125** are etched using the patterned photoresist layer **142** as a template to form the first gate mask **122** and expose portions of the gate electrode layer **115** adjacent the first gate mask **122**. In some embodiments, the photoresist layer **140** remains intact over the second active region **102B**, so as to isolate the processing of the second active region **102B** from that of the first active region **102A**.

(15) Referring to FIG. 3, an etching process is performed to pattern the gate electrode layer **115** over the first active region **102A** using the first gate mask **122** as an etch template to define a first gate electrode **145** over the first active region **102A**, in accordance with some embodiments. In some embodiments, the etching process is at least one of a plasma etching process, a reactive ion etching (RIE) process, or other applicable techniques. In some embodiments, the gate dielectric layer **110** is not etched when etching the gate electrode layer **115**. In some embodiments, the etching process also patterns the gate dielectric layer **110** as indicated by the dashed lines. In some embodiments, the chemistry of the etching process may be changed from a first etch chemistry for etching the gate electrode layer **115** and a second etch chemistry for etching the gate dielectric layer **110**.

(16) Referring to FIG. 4, an implantation process **150** is performed using the first gate mask **122** as an implantation template to implant a first dopant into the first active region **102A** of the semiconductor layer **105** and form a first doped region **155** in the semiconductor layer **105** in the first active region **102A**, in accordance with some embodiments. In some embodiments, the first doped region **155** is a lightly doped drain (LDD) region having a first dopant concentration. In some embodiments, the first doped region **155** has a depth **157** greater than a height **160** of the first gate electrode **145** as measured from a top surface of the semiconductor layer **105**. Using the first gate mask **122** as the implantation template allows the first doped region **155** to be formed with a greater depth **157** without introducing dopant into the gate dielectric layer **110**, which can mitigate hot carrier injection and thereby improve hot carrier injection performance issues. In some embodiments, the greater depth **157** of the first doped region **155** results from the use of a higher implantation energy. The use of the first gate mask **122** as an implantation template allows a complete retrograde junction to be formed due to the higher implantation energy. In some embodiments, the implantation process **150** is a vertical implantation process. Compared to angled implantation processes for forming LDD regions, the use of a vertical implantation process reduces threshold voltage mismatches. In some embodiments, the first mask layer **120** in place over the second active region **102B** isolates the second active region **102B** from the implantation process **150**.

(17) In some embodiments, such as where the resulting transistor is an n-type metal-oxide-semiconductor (nMOS) device, the first doped region **155** comprises an n-type impurity, such as phosphorous, arsenic, or a different n-type dopant. In some embodiments, such as where the resulting transistor is a p-type metal-oxide-semiconductor (pMOS) device, the first doped region **155** comprises a p-type impurity such as boron, BF₂, or a different p-type dopant. In some embodiments, the implantation process **150** is performed through the gate dielectric layer **110** to protect an upper surface of the semiconductor layer **105** from implantation damage.

(18) Referring to FIG. 5, an etch process is performed to remove exposed portions of the gate dielectric layer **110** and removal processes, such as ashing, etching, etc., are performed to remove the photoresist layer **140** from over the second active region **102B** and to remove the patterned photoresist layer **142**, the OPL layer **135**, and the BARC layer **130** from over the first active region **102A**, in accordance with some embodiments. The first gate electrode **145** and the remaining portion of the gate dielectric layer **110** define a first gate structure **165**. In some embodiments, the first gate structure **165** is a sacrificial, or “dummy” gate structure, which is later removed and replaced with a replacement gate structure comprising a high-k gate dielectric layer and a metal gate electrode.

(19) Referring to FIG. 6, a sidewall spacer **170** is formed adjacent the first gate structure **165**, in accordance with some embodiments. In some embodiments, the sidewall spacer **170** is formed by depositing a conformal spacer layer over the first gate structure **165** and performing an anisotropic etch process to remove portions of the spacer layer positioned on horizontal surfaces of the hard mask layer **125** and the semiconductor layer **105**. In some embodiments, the sidewall spacer **170** comprises the same material as the hard mask layer. In some embodiments, the sidewall spacer **170** comprises nitrogen and silicon. For ease of illustration and to avoid obscuring the first doped region **155**, the sidewall spacer **170** is not illustrated in the plan view.

(20) Referring to FIG. 7, an implantation process **175** is performed using the first gate structure **165** and the sidewall spacer **170** as an implantation template to implant a second dopant into the semiconductor layer **105** and form a second doped region **180** in the first active region **102A** of the semiconductor layer **105**, in accordance with some embodiments. In some embodiments, the second doped region **180** is formed within the first doped region **155** such that the first doped region **155** surrounds the second doped region **180**. In some embodiments, the second doped region **180** is a highly doped source/drain (S/D) region having a second dopant concentration greater than the first dopant concentration of the first doped region **155**. In some embodiments, the second

doped region **180** has a depth **185** less than the depth **157** of the first doped region **155** as measured from a top surface of the semiconductor layer **105**. In some embodiments, the first dopant and the second dopant are the same. In some embodiments, the first mask layer **120** in place over the second active region **102B** isolates the second active region **102B** from the implantation process **175**.

(21) Turning to FIG. **8**, in some embodiments, the implantation process **150** to form the first doped region **155**, as described with respect to FIG. **4**, is performed prior to etching the gate electrode layer **115**, as described with respect to FIG. **3**. During the implantation process **150** dopant is introduced into the semiconductor layer **105** through the portions of the gate electrode layer **115** and the gate dielectric layer **110** not covered by the first gate mask **122**. In some embodiments, the energy of the implantation process **150** is configured to account for the presence of the gate electrode layer **115** and the gate dielectric layer **110**. In some embodiments, the first doped region **155** has a depth **157** greater than a height **160** of the first gate electrode **145**. Using the first gate mask **122** as the implantation template allows the first doped region **155** to be formed with a greater depth **157** without introducing dopant into the gate dielectric layer **110**, which can lead to hot carrier injection performance issues. In some embodiments, the first mask layer **120** in place over the second active region **102B** isolates the second active region **102B** from the implantation process **150**.

(22) Referring to FIG. **9**, the second hard mask layer **137** is removed from over the second active region **102B** and a photoresist layer **195** is formed over the semiconductor arrangement **100** after the implantation process **175** regardless of whether the first doped region **155** is formed prior to the etching of the gate electrode layer **115**, as described with respect to FIG. **8**, or after the etching of the gate electrode layer **115**, as described with respect to FIG. **4**. In some embodiments, an etch process is performed to remove the second hard mask layer **137**. In some embodiments, the combination of one or more of the photoresist layer **195**, the OPL layer **135**, the BARC layer **130**, and the hard mask layer **125** formed over the second active region **102B** may be referred to as a second mask layer **200**.

(23) Referring to FIG. **10**, the second mask layer **200** is patterned to define a second gate mask **202**, in accordance with some embodiments. In some embodiments, the photoresist layer **195** is exposed using a radiation source and a reticle to define a pattern in the photoresist layer **195**. In some embodiments, portions of the photoresist layer **195** are removed to define a patterned photoresist layer **197**. In some embodiments, the underlying OPL layer **135**, BARC layer **130**, and hard mask layer **125** are etched using the patterned photoresist layer **197** as a template to form the second gate mask **202** and expose portions of the gate electrode layer **115** adjacent the second gate mask **202**. In some embodiments, the photoresist layer **195** remains intact over the first active region **102A**, so as to isolate the processing of the second active region **102B** from that of the first active region **102A**.

(24) In some embodiments, due to manufacturing tolerances, the second gate mask **202** overlying the second active region **102B** is not perfectly aligned with the first gate structure **165** overlying the first active region **10A**, where a perfect alignment would result in a first sidewall of the second gate mask **202** being co-planar with a first sidewall of the first gate electrode **145** and a second sidewall of the second gate mask **202** being co-planar with a second sidewall of the first gate electrode **145**. Thus, in some embodiments, there is misalignment between the second gate mask **202** and the first gate electrode **145**, such that the sidewalls of the second gate mask **202** are not co-planar with the sidewalls of the first gate electrode **145**. As will be described further with respect to FIG. **17**, in some embodiments, this misalignment between the second gate mask **202** and the first gate electrode **145** results in alignment variation between the first gate structure **165** and a second gate structure **810B**, which is formed using the second gate mask **202**.

(25) Referring to FIG. **11**, an etching processes is performed to pattern the gate electrode layer **115** over the second active region **102B** using the second gate mask **202** as an etch template to define a second gate electrode **205** over the second active region **102B**, in accordance with some

embodiments. In some embodiments, the etching process is at least one of a plasma etching process, a reactive ion etching (RIE) process, or other applicable techniques. In some embodiments, the gate dielectric layer **110** is not etched when etching the gate electrode layer **115**. In some embodiments, the etching process also patterns the gate dielectric layer **110** as indicated by the dashed lines. In some embodiments, the chemistry of the etching process may be changed from a first etch chemistry for etching the gate electrode layer **115** and a second etch chemistry for etching the gate dielectric layer **110**.

(26) Referring to FIG. **12**, an implantation process **210** is performed using the second gate mask **202** as an implantation template to implant a third dopant into the second active region **102B** of the semiconductor layer **105** and form a third doped region **215** in the semiconductor layer **105** in the second active region **102B**, in accordance with some embodiments. In some embodiments, the third doped region **215** is a lightly doped drain (LDD) region having a third dopant concentration. In some embodiments, the third doped region **215** has a depth **220** greater than a height **224** of the second gate electrode **205** as measured from a top surface of the semiconductor layer **105**. Using the second gate mask **202** as the implantation template allows the third doped region **215** to be formed with a greater depth **220** without introducing dopant into the gate dielectric layer **110**, which can mitigate hot carrier injection and thereby improve hot carrier injection performance issues. In some embodiments, the greater depth **220** of the third doped region **215** results from the use of a higher implantation energy. The use of the second gate mask **202** as an implantation template allows a complete retrograde junction to be formed due to the higher implantation energy. In some embodiments, the implantation process **210** is a vertical implantation process. Compared to angled implantation processes for forming LDD regions, the use of a vertical implantation process reduces threshold voltage mismatches. In some embodiments, the photoresist layer **195** in place over the first active region **102A** isolates the first active region **102A** from the implantation process **210**.

(27) In some embodiments, such as where the resulting transistor is an n-type metal-oxide-semiconductor (nMOS) device, the third doped region **215** comprises an n-type impurity, such as phosphorous, arsenic, or a different n-type dopant. In some embodiments, such as where the resulting transistor is a p-type metal-oxide-semiconductor (pMOS) device, the third doped region **215** comprises a p-type impurity such as boron, BF₂, or a different p-type dopant. In some embodiments, the implantation process **210** is performed through the gate dielectric layer **110** to protect an upper surface of the semiconductor layer **105** from implantation damage. In some embodiments, when the first doped region **155** comprises a p-type dopant, the third doped region **215** comprises an n-typed dopant. In some embodiments, when the first doped region **155** comprises an n-type dopant, the third doped region **215** comprises a p-typed dopant.

(28) Referring to FIG. **13**, an etch process is performed to remove exposed portions of the gate dielectric layer **110** and removal processes, such as ashing, etching, etc., are performed to remove the photoresist layer **195** from over the first active region **102A** and to remove the patterned photoresist layer **197**, the OPL layer **135**, and the BARC layer **130** from over the second active region **102B**, in accordance with some embodiments. The second gate electrode **205** and the remaining portion of the gate dielectric layer **110** define a second gate structure **225**. In some embodiments, the second gate structure **225** is a sacrificial, or “dummy” gate structure, which is later removed and replaced with a replacement gate structure comprising a high-k gate dielectric layer and a metal gate electrode.

(29) Referring to FIG. **14**, a sidewall spacer **235** is formed adjacent the second gate structure **225**, in accordance with some embodiments. In some embodiments, the sidewall spacer **235** is formed by depositing a conformal spacer layer over the second gate structure **225** and performing an anisotropic etch process to remove portions of the spacer layer positioned on horizontal surfaces of the hard mask layer **125** and the semiconductor layer **105**. In some embodiments, the sidewall spacer **235** comprises the same material as the hard mask layer. In some embodiments, the sidewall

spacer **235** comprises nitrogen and silicon. For ease of illustration and to avoid obscuring the third doped region **215**, the sidewall spacer **235** is not illustrated in the plan view. In some embodiments, the sidewall spacer **235** is also formed adjacent the first gate structure **165**. In some embodiments, the sidewall spacer **235** comprises a different material than the sidewall spacer **170** to allow selective removal of the sidewall spacer **235** adjacent the first gate structure **165** using masking and etching processes.

(30) Referring to FIG. **15**, a mask layer **240** is formed above the first active region **102A** and the first gate structure **165**, and an implantation process **245** is performed using the second gate structure **225** and the sidewall spacer **170** as an implantation template to implant a fourth dopant into the semiconductor layer **105** and form a fourth doped region **250** in the second active region **102B** of the semiconductor layer **105**, in accordance with some embodiments. In some embodiments, the mask layer **240** comprises a photoresist layer. In some embodiments, the fourth doped region **250** is formed within the third doped region **215** such that the third doped region **215** surrounds the fourth doped region **250**. In some embodiments, the fourth doped region **250** is a highly doped source/drain (S/D) region having a fourth dopant concentration greater than the third dopant concentration of the third doped region **215**. In some embodiments, the fourth doped region **250** has a depth **255** less than the depth **220** of the third doped region **215** as measured from a top surface of the semiconductor layer **105**. In some embodiments, the third dopant and the fourth dopant are the same. In some embodiments, the mask layer **240** in place over the first active region **102A** isolates the first active region **102A** from the implantation process **245**.

(31) Referring to FIG. **16**, the mask layer **240** is removed, in accordance with some embodiments. In some embodiments, an etch process or an ashing process is performed to remove the mask layer **240**.

(32) As described herein, the first gate structure **165** and the first doped region **155** are formed prior to forming the second gate structure **225** and the third doped region **215**. In some embodiments, the order is reversed, such that the second gate structure **225** and the third doped region **215** are formed first. Since the first gate structure **165** and the second gate structure **225** are formed using separate process flows, an interface **270** is defined where the first gate structure **165** contacts the second gate structure **225**. In some embodiments, the interface **270** is defined wherein the first gate electrode **145** of the first gate structure **165** contacts the second gate electrode **205** of the second gate structure **225**. Moreover, some degree of alignment variation may exist between the first gate structure **165** and the second gate structure **225**. In some embodiments, a first long axis **275A** is defined along a center of the first gate structure **165**, and a second long axis **275B** is defined along a center of the second gate structure **225**. In some embodiments, the first long axis **275A** is parallel to the second long axis **275B**. Due to the alignment variation, the first long axis **275A** is laterally offset with respect to the second long axis **275B**.

(33) In some embodiments, the first active region **102A** is associated with a first transistor **280A**, and the second active region **102B** is associated with a second transistor **280B**. The first gate structure **165** and the second gate structure **225** define a shared gate structure for the first transistor **280A** and the second transistor **280B** because the first gate structure **165** is electrically coupled to the second gate structure **225** due to the first gate structure **165** contacting the second gate structure **225** at the interface **270**. In some embodiments, the dopant used to form the first doped region **155** of the first transistor **280A** has a first dopant type, and the dopant used to form the third doped region **215** of the second transistor **280B** has a second dopant type different from the first dopant type. In some embodiments, the first dopant type comprises an n-type dopant and the second dopant type comprises a p-type dopant. In some embodiments, the first transistor **280A** is an nMOS transistor and the second transistor **280B** is a pMOS transistor. In some embodiments, the first transistor **280A** and the second transistor **280B** define an inverter circuit in a static random access memory (SRAM) cell having a shared gate structure defined by the first gate structure **165** and the second gate structure **225**.

(34) According to some embodiments, the use of the first gate mask **122** as an etch template for forming the first gate electrode **145**, and the use of the first gate mask **122** and the first gate electrode **145** as an implantation template facilitates the formation of the first doped region **155** with an increased depth resulting from an increased implantation energy. The first doped region **155** is formed at the increased depth without introducing dopant into the gate dielectric layer **110** and introducing hot carrier injection defects. In some embodiments, the implantation process **150** is a vertical implantation process. Compared to angled implantation processes for forming LDD regions, the use of a vertical implantation process reduces threshold voltage mismatches. In some embodiments, the use of the second gate mask **202** as an etch template for forming the second gate electrode **205**, and the use of the second gate mask **202** and the gate electrode **210** as an implantation template provides similar advantages for the third doped region **215**.

(35) In some embodiments, a method of forming a semiconductor arrangement includes forming a gate dielectric layer over a semiconductor layer. A gate electrode layer is formed over the gate dielectric layer. A first gate mask is formed over the gate electrode layer. The gate electrode layer is etched using the first gate mask as an etch template to form a first gate electrode. A first dopant is implanted into the semiconductor layer using the first gate mask and the first gate electrode as an implantation template to form a first doped region in the semiconductor layer.

(36) In some embodiments, a method of forming a semiconductor arrangement includes forming a gate dielectric layer over a semiconductor layer. A gate electrode layer is formed over the gate dielectric layer. A first gate mask is formed over the gate electrode layer. The gate electrode layer is etched using the first gate mask as an etch template to form a first gate electrode. A first dopant is implanted into the semiconductor layer using the first gate mask as an implantation template to form a first doped region in the semiconductor layer. The first doped region extends to a first depth below a top surface of the semiconductor layer. A first sidewall spacer is formed adjacent the first gate electrode after implanting the first dopant. A second dopant is implanted using the first sidewall spacer and the first gate electrode as an implantation template to form a second doped region in the semiconductor layer. The second doped region extends to a second depth below the top surface of the semiconductor layer. The second depth is less than the first depth.

(37) In some embodiments, a semiconductor arrangement includes a first transistor including a first gate structure over a semiconductor layer and a first doped region in the semiconductor layer adjacent the first gate structure. A second transistor includes a second gate structure over the semiconductor layer and a second doped region in the semiconductor layer adjacent the second gate structure. The second gate structure contacts the first gate structure such that an interface is defined between the first gate structure and the second gate structure.

(38) The foregoing outlines features of several embodiments so that those of ordinary skill in the art may better understand various aspects of the present disclosure. Those of ordinary skill in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of various embodiments introduced herein. Those of ordinary skill in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

(39) Although the subject matter has been described in language specific to structural features or methodological acts, it is to be understood that the subject matter of the appended claims is not necessarily limited to the specific features or acts described above. Rather, the specific features and acts described above are disclosed as example forms of implementing at least some of the claims.

(40) Various operations of embodiments are provided herein. The order in which some or all of the operations are described should not be construed to imply that these operations are necessarily order dependent. Alternative ordering will be appreciated having the benefit of this description. Further, it will be understood that not all operations are necessarily present in each embodiment

provided herein. Also, it will be understood that not all operations are necessary in some embodiments.

(41) It will be appreciated that layers, features, elements, etc. depicted herein are illustrated with particular dimensions relative to one another, such as structural dimensions or orientations, for example, for purposes of simplicity and ease of understanding and that actual dimensions of the same differ substantially from that illustrated herein, in some embodiments. Additionally, a variety of techniques exist for forming the layers, regions, features, elements, etc. mentioned herein, such as at least one of etching techniques, planarization techniques, implanting techniques, doping techniques, spin-on techniques, sputtering techniques, growth techniques, or deposition techniques such as chemical vapor deposition (CVD), for example.

(42) Moreover, “exemplary” is used herein to mean serving as an example, instance, illustration, etc., and not necessarily as advantageous. As used in this application, “or” is intended to mean an inclusive “or” rather than an exclusive “or”. In addition, “a” and “an” as used in this application and the appended claims are generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form. Also, at least one of A and B and/or the like generally means A or B or both A and B. Furthermore, to the extent that “includes”, “having”, “has”, “with”, or variants thereof are used, such terms are intended to be inclusive in a manner similar to the term “comprising”. Also, unless specified otherwise, “first,” “second,” or the like are not intended to imply a temporal aspect, a spatial aspect, an ordering, etc. Rather, such terms are merely used as identifiers, names, etc. for features, elements, items, etc. For example, a first element and a second element generally correspond to element A and element B or two different or two identical elements or the same element.

(43) Also, although the disclosure has been shown and described with respect to one or more implementations, equivalent alterations and modifications will occur to others of ordinary skill in the art based upon a reading and understanding of this specification and the annexed drawings. The disclosure comprises all such modifications and alterations and is limited only by the scope of the following claims. In particular regard to the various functions performed by the above described components (e.g., elements, resources, etc.), the terms used to describe such components are intended to correspond, unless otherwise indicated, to any component which performs the specified function of the described component (e.g., that is functionally equivalent), even though not structurally equivalent to the disclosed structure. In addition, while a particular feature of the disclosure may have been disclosed with respect to only one of several implementations, such feature may be combined with one or more other features of the other implementations as may be desired and advantageous for any given or particular application.

Claims

1. A semiconductor arrangement, comprising: a first transistor, comprising: a first source/drain region; a second source/drain region; a first gate electrode between the first source/drain region and the second source/drain region; a first sidewall spacer; and a second sidewall spacer, wherein the second sidewall spacer is separated from the first gate electrode by the first sidewall spacer; and a second transistor, comprising: a second gate electrode contacting the first gate electrode at a sidewall extending in a direction parallel to a direction extending from the first source/drain region to the second source/drain region; and the second sidewall spacer, wherein the second sidewall spacer is in contact with the second gate electrode.
2. The semiconductor arrangement of claim 1, wherein the second gate electrode contacts the first gate electrode such that an interface is defined between the first gate electrode and the second gate electrode.
3. The semiconductor arrangement of claim 2, wherein a center of the second gate electrode is laterally offset from a center of the first gate electrode at the interface.

4. The semiconductor arrangement of claim 1, wherein: the first transistor comprises a first doped region, the first source/drain region is a second doped region over the first doped region, and the first doped region underlies the first sidewall spacer and the second sidewall spacer.
5. The semiconductor arrangement of claim 4, wherein the second doped region underlies the second sidewall spacer.
6. The semiconductor arrangement of claim 1, wherein: the first sidewall spacer has a first material composition, and the second sidewall spacer has a second material composition different than the first material composition.
7. The semiconductor arrangement of claim 1, wherein the second transistor is a p-type metal-oxide semiconductor (pMOS) device and the first transistor is an n-type metal-oxide semiconductor (nMOS) device.
8. A semiconductor arrangement, comprising: a first transistor comprising: a first source/drain region; a second source/drain region; and a first gate structure comprising a first sidewall spacer having a first material composition, wherein the first gate structure is between the first source/drain region and the second source/drain region; and a second transistor comprising a second gate structure, wherein: the second gate structure contacts the first gate structure at a sidewall extending in a direction parallel to a direction extending from the first source/drain region to the second source/drain region such that an interface is defined between the first gate structure and the second gate structure, and the second gate structure comprises a second sidewall spacer having a second material composition different than the first material composition.
9. The semiconductor arrangement of claim 8, wherein a center of the second gate structure is laterally offset from a center of the first gate structure at the interface.
10. The semiconductor arrangement of claim 8, wherein: the first gate structure comprises a first gate electrode, the first sidewall spacer is in contact with the first gate electrode, the second gate structure comprises a second gate electrode, and the second sidewall spacer is in contact with the second gate electrode.
11. The semiconductor arrangement of claim 10, wherein the first transistor comprises the second sidewall spacer.
12. The semiconductor arrangement of claim 8, wherein the first sidewall spacer comprises silicon and nitrogen.
13. The semiconductor arrangement of claim 8, wherein the second transistor is a p-type metal-oxide semiconductor (pMOS) device and the first transistor is an n-type metal-oxide semiconductor (nMOS) device.
14. A method comprising: forming a gate electrode layer; forming a first photoresist layer over the gate electrode layer; removing a first portion of the first photoresist layer to define a patterned first photoresist layer; removing a first portion of the gate electrode layer not underlying the patterned first photoresist layer to define a first gate electrode from a second portion the gate electrode layer; removing the patterned first photoresist layer after defining the first gate electrode; forming a second photoresist layer over first gate electrode and a third portion of the gate electrode layer; removing a first portion of the second photoresist layer to define a patterned second photoresist layer; and removing a first part of the third portion of the gate electrode layer not underlying the patterned second photoresist layer to define a second gate electrode from a second part of the third portion of the gate electrode layer.
15. The method of claim 14, comprising: performing an implantation process to implant a first dopant into a substrate after removing the first portion of the gate electrode layer and before removing the patterned first photoresist layer.
16. The method of claim 14, comprising: forming a first sidewall spacer adjacent the first gate electrode before forming the second photoresist layer.
17. The method of claim 16, comprising: performing a first implantation process to implant a first dopant into a substrate after removing the first portion of the gate electrode layer and before

removing the patterned first photoresist layer; and performing a second implantation process to implant a second dopant into the substrate after forming the first sidewall spacer and before forming the second photoresist layer.

18. The method of claim 14, comprising: forming a first sidewall spacer adjacent the first gate electrode before forming the second photoresist layer; and forming a second sidewall spacer adjacent the first sidewall spacer after removing the first part of the third portion of the gate electrode layer.

19. The method of claim 14, comprising: forming a sidewall spacer around the first gate electrode after removing the first part of the third portion of the gate electrode layer.

20. The semiconductor arrangement of claim 8, wherein: the first transistor comprises a first doped region, the first source/drain region is a second doped region over the first doped region, and the first doped region underlies the first sidewall spacer.
